

الروشتة ف الإحصاء

Lecture 1 - Summation basics

Sample space in the lecture example

$$S = \{1, 2, 3, 4, 5, 6, 7\}$$

Sum of observations

$$\sum_{i=1}^n X_i = X_1 + X_2 + \dots + X_n$$

Sum of squares

$$\sum_{i=1}^n X_i^2 = X_1^2 + X_2^2 + \dots + X_n^2$$

Square of the total sum

$$\left(\sum_{i=1}^n X_i \right)^2$$

Adding or subtracting a constant inside sigma

$$\sum_{i=1}^n (X_i - c) = \sum_{i=1}^n X_i - nc$$

Multiplying by a constant

$$\sum_{i=1}^n aX_i = a \sum_{i=1}^n X_i$$

Linear form

$$\sum_{i=1}^n (aX_i + b) = a \sum_{i=1}^n X_i + nb$$

Division by a constant

$$\sum_{i=1}^n \frac{X_i}{a} = \frac{1}{a} \sum_{i=1}^n X_i$$

Lecture 2 - Measures of central tendency

Arithmetic mean for ungrouped data

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

Arithmetic mean for grouped data

$$\bar{X} = \frac{\sum_{i=1}^m F_i X_i}{\sum_{i=1}^m F_i}$$

Geometric mean

$$GM = \sqrt[n]{\prod_{i=1}^n X_i} = \left(\prod_{i=1}^n X_i \right)^{1/n}$$

Harmonic mean

$$HM = \frac{n}{\sum_{i=1}^n \frac{1}{X_i}}$$

Finding a missing value from the mean

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

Lecture 3 - Median and Mode

Median position for odd n

$$r_{\text{median}} = \frac{n + 1}{2}$$

Median positions for even n

$$r_1 = \frac{n}{2}, \quad r_2 = \frac{n}{2} + 1$$

Median for grouped data

$$\text{Median} = L + \left[\frac{n/2 - C}{F} \right] h$$

Mode for grouped data

$$\text{Mode} = L + \left[\frac{F_1 - F_0}{2F_1 - F_0 - F_2} \right] h$$

Lecture 4 - Quartiles and dispersion

First quartile for odd n

$$Q_1 = \frac{n + 1}{4}$$

First quartile for even n

$$Q_1 = \left(\frac{n}{4}, \frac{n}{4} + 1 \right)$$

Second quartile / median for odd n

$$Q_2 = \frac{n + 1}{2}$$

Second quartile / median for even n

$$Q_2 = \left(\frac{n}{2}, \frac{n}{2} + 1 \right)$$

Third quartile for odd n

$$Q_3 = \frac{3(n + 1)}{4}$$

Third quartile for even n

$$Q_3 = \left(\frac{3n}{4}, \frac{3n}{4} + 1 \right)$$

Quartile formula for grouped data

$$Q = L + \left[\frac{\text{position} - C}{F} \right] h$$

Interquartile range

$$IR = Q_3 - Q_1$$

Semi-interquartile range

$$SIR = \frac{1}{2}IR$$

Range

$$Range = Max - Min$$

Lecture 5 - Measures of variability

Variance for ungrouped data

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2$$

Standard deviation

$$\sigma = \sqrt{\sigma^2}$$

Variance for grouped data

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^m F_i (X_i - \bar{X})^2 = \frac{1}{n} \sum_{i=1}^m F_i X_i^2 - \bar{X}^2$$

Coefficient of variation

$$C.V = \frac{\sigma}{\mu} \times 100$$

Lecture 6 - Skewness and mean deviation

Pearson skewness using median

$$Sk = \frac{3(\bar{X} - Median)}{\sigma}$$

Pearson skewness using mode

$$Sk = \frac{\bar{X} - Mode}{\sigma}$$

Mean deviation

$$MD = \frac{1}{n} \sum_{i=1}^n |X_i - \bar{X}|$$

Direction of skewness

$$Sk > 0, \quad Sk = 0, \quad Sk < 0$$

Lecture 7 - Moments and kurtosis

First moment about origin

$$m_1(0) = \frac{1}{n} \sum_{i=1}^n x_i$$

Second moment about origin

$$m_2(0) = \frac{1}{n} \sum_{i=1}^n x_i^2$$

Third moment about origin

$$m_3(0) = \frac{1}{n} \sum_{i=1}^n x_i^3$$

Fourth moment about origin

$$m_4(0) = \frac{1}{n} \sum_{i=1}^n x_i^4$$

r-th moment about origin

$$m_r(0) = \frac{1}{n} \sum_{i=1}^n x_i^r$$

First and second central moments

$$m_1 = 0, \quad m_2 = m_2(0) - [m_1(0)]^2 = \sigma^2$$

Third central moment

$$m_3 = m_3(0) - 3m_1(0)m_2(0) + 2[m_1(0)]^3$$

Fourth central moment

$$m_4 = m_4(0) - 4m_1(0)m_3(0) + 6[m_1(0)]^2m_2(0) - 3[m_1(0)]^4$$

Moment skewness coefficient

$$Sk = \frac{m_3}{\sigma^3}$$

Kurtosis coefficient

$$Ku = \frac{m_4}{\sigma^4}$$

Relation between kurtosis and skewness

$$Ku \geq Sk^2 + 1$$

Lecture 8 - Correlation

Pearson correlation coefficient

$$r_{xy} = \frac{n\sum x_i y_i - (\sum x_i)(\sum y_i)}{\sqrt{n\sum x_i^2 - (\sum x_i)^2} \sqrt{n\sum y_i^2 - (\sum y_i)^2}}$$

Sample means

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

Covariance

$$C_{xy} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

Degree of correlation

r = 0	0 < r < 0.5	r = 0.5	0.5 < r < 1	r = 1
No correlation	Weak	Moderate	Strong	Perfect

Types of correlation

- If r is positive (+), the relation between X and Y is direct (positive) correlation.
- If r is negative (-), the relation between X and Y is inverse (negative) correlation.

Lecture 8 - Spearman rank correlation

Spearman rank correlation formula

$$r_s = 1 - \frac{6\sum d_i^2}{n(n^2 - 1)}$$

Where

- d = difference between Rank X and Rank Y for each pair of observations.
- $\sum d^2$ = sum of squared rank differences.
- n = number of paired observations.

Interpretation

- If $r_s > 0$, the correlation is direct.
- If $r_s < 0$, the correlation is inverse.
- The closer $|r_s|$ is to 1, the stronger the correlation.